

Test and Measurement For WIRELESS APPLICATIONS For SHIPS And SUBMARINES

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This article discusses test and measurement for wireless applications for ships and submarines

Platform-based ship and submarine systems used for naval applications have various systems for different applications, including communication, radar, signal monitoring and electronic warfare. These systems operate in different frequency bands operating in audio, radio frequency (RF) and microwave ranges, covering a broad spectrum from a few Hz to GHz. Communication with submarines is difficult because radio waves do not travel well through the deep sea.

Several techniques are used to communicate with submarines. Underwater acoustic communication is used for sending and receiving messages under water. There are several ways of achieving such communication, but the most common is using hydrophones.

Very low frequency (VLF) radio waves, ranging from 3kHz to 30kHz, can penetrate seawater to depths of approximately 20 metres. Submarines that are more deeply submerged might use a buoy equipped with an antenna on a long cable. A buoy is usually raised a few metres below the surface.

A surfaced submarine can use ordinary radio communication on higher frequencies, such as high frequency (3MHz - 30MHz), very high frequency (30MHz - 300MHz) or ultra high frequency (300MHz to 3GHz), and

transmit information via voice.

Srisailam Todeti, senior applications specialist, Keysight Technologies India, says "Underwater communication is difficult due to a number of factors, such as time variations of the channel, multipath propagation, strong signal attenuation and small available bandwidth. Thus, development of communication systems requires rigorous testing to ensure that these can operate under demanding environments.

"For true evaluation of the device under test (DUT), the selected test instrument should maintain high accuracy and performance against different environmental conditions. Hence, a standardised instrument can help with a better analysis.

"If required, the same test instrument can be used in systems, in place of some sub-systems. Before going to field, these systems can be tested by injecting the signals with different signal impairments in a lab. Calibration of the test setup is required to inject corrected signals into the DUT."

Test solutions for wireless applications under the deep sea

To test and maintain the above-mentioned systems in the field, instruments such as function/arbitrary waveform generator, audio analyser, signal generator, spectrum analyser, vector network analyser, channel emulator and so on are used. Then, there are software available that help in generating and analysing custom signals.

Analogue and vector signal generators generate signals such as AM, FM, pulse, QPSK, QAM, chirp, barker modulation, antenna scan pattern and the like.

A vector network analyser is used for component characterisation, and cable and antenna testing.

SystemVue software for radar signal generation is used to generate different kinds of waveforms and to download the signals into

Fig. 1: MS2850A
(Credit: Anritsu India Pvt Ltd)

